Sequential Conditional Means of Equicorrelated Scores

A fixed starting score and the average of subsequent predictions

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Fix a starting standardized score a. For jointly Gaussian tests with equal pairwise correlation c, recursively predict each subsequent test by conditioning on the starting score and on all earlier predictions. Every prediction equals ca, so the average of any positive number of these predictions is ca. The starting score is excluded from that average. We derive the Gaussian conditional mean, solve the recursion, describe the admissible correlations and singular endpoints, and distinguish this deterministic construction from actual future scores. A counterexample shows that means, variances and correlations alone do not establish the result.

AI disclosure. This analysis, its mathematical exposition, supporting repository documentation and typesetting were prepared with OpenAI Codex and collaborating AI agents. The requester specified the problem, clarified the indexing and requested this analysis. Mathematical review and build checks were performed within the same AI-assisted workflow; no independent human peer review or new empirical psychometric validation is claimed.

1 Problem and notation

Let $k \geq 1$ be an integer and write

$$X_0, X_1, \dots, X_k$$

for standardized test scores, each with mean zero and variance one. We relabel the original tests $A_r(c, 0, 1)$ so that test 0 supplies the starting observation. For distinct indices, assume $\operatorname{Cov}(X_i, X_j) = \operatorname{Corr}(X_i, X_j) = c$.

Use $a \in \mathbb{R}$ for the *fixed starting score*; it is a scalar, not a function. For each $1 \leq n \leq k$, let g_n be the Gaussian conditional mean function of X_n given $(X_0, \dots, X_{n-1})$. Define

$$e_1 = g_1(a) = \mathrm{E}[X_1 \mid X_0 = a], \tag{1}$$

$$\begin{aligned} e_n &= g_n(a, e_1, \dots, e_{n-1}) \\ &= \mathrm{E}[X_n \mid X_0 = a, X_1 = e_1, \dots, X_{n-1} = e_{n-1}], \qquad 2 \leq n \leq k. \end{aligned} \tag{2}$$

The quantity of interest is

$$\boxed{\bar{e}_k = \frac{1}{k} \sum_{n=1}^{k} e_n.} \tag{3}$$

It contains the k predictions $e_1, \dots, e_k$, and excludes a.

For continuous distributions, conditioning on a particular score is not a ratio of probabilities of point events. A conditional mean is ordinarily defined only almost everywhere. Here, g_n means the continuous version supplied by Gaussian regression, derived below. This convention specifies the recursion even at individual histories of probability zero. The singular case $c = 1$ will be treated directly on its support.

2 The joint model and its correlation domain

Assume $(X_0, \dots, X_k)$ is jointly Gaussian, allowing degenerate Gaussian distributions at the endpoints. Its covariance matrix is

$$R_{k+1} = (1-c)I_{k+1} + c\mathbf{1}_{k+1}\mathbf{1}_{k+1}^{\mathsf{T}}, \tag{4}$$

where I_d is the d-dimensional identity and $\mathbf{1}_d$ is a column of d ones. Vectors orthogonal to $\mathbf{1}_{k+1}$ have eigenvalue $1-c$; the vector $\mathbf{1}_{k+1}$ has eigenvalue $1+kc$. Thus a covariance matrix for these $k+1$ scores exists exactly when

$$\boxed{-\frac{1}{k} \le c \le 1.} \tag{5}$$

An arbitrarily long equicorrelated collection requires $0 \le c \le 1$. A negative fixed c cannot satisfy the lower bound for all k.

For $0 \le c \le 1$, a construction consistent with the hypothetical common-factor illustration in the repository's Hoeflin/Ferguson analysis, Appendix A, is

$$X_j = \sqrt{c}\,G + \sqrt{1-c}\,\varepsilon_j, \tag{6}$$

where $G, \varepsilon_0, \dots, \varepsilon_k$ are independent standard-normal variables. The covariance-matrix derivation below also covers the admissible negative correlations. This is a mathematical model, not a claim that actual test batteries satisfy these assumptions.

3 Derivation of the conditional mean

Fix $1 \le n \le k$, initially with $c < 1$, and put

$$Y_n = (X_0, \dots, X_{n-1})^{\mathsf{T}}, \qquad R_n = \mathrm{Cov}(Y_n) = (1-c)I_n + c\mathbf{1}_n\mathbf{1}_n^{\mathsf{T}}.$$

The eigenvalue of R_n in the direction $\mathbf{1}_n$ is

$$d_n = 1 + (n-1)c.$$

It is positive throughout $-1/k \le c < 1$: for negative c, $d_n \ge 1 - (n-1)/k \ge 1/k > 0$. The other eigenvalues are $1-c > 0$. Consequently every conditioning block is nonsingular, even if the full $(k+1)$-score distribution is singular at $c = -1/k$.

An explicit inverse, verified by multiplying it by R_n, is

$$R_n^{-1} = \frac{1}{1-c}\left(I_n - \frac{c}{1+(n-1)c}\mathbf{1}_n\mathbf{1}_n^{\mathsf{T}}\right). \tag{7}$$

Since $\mathrm{Cov}(X_n, Y_n) = c\mathbf{1}_n^{\mathsf{T}}$, the regression coefficient is

$$\mathrm{Cov}(X_n, Y_n)R_n^{-1} = \frac{c}{1+(n-1)c}\mathbf{1}_n^{\mathsf{T}}. \tag{8}$$

To establish why this coefficient gives the conditional mean, define the residual

$$W_n = X_n - \frac{c}{1+(n-1)c}\sum_{j=0}^{n-1} X_j. \tag{9}$$

For each $\ell \in \{0, \dots, n-1\}$,

$$\begin{aligned}\operatorname{Cov}(W_n, X_\ell) &= c - \frac{c}{1+(n-1)c}\sum_{j=0}^{n-1}\operatorname{Cov}(X_j, X_\ell)\\ &= c - \frac{c}{1+(n-1)c}[1+(n-1)c] = 0.\end{aligned}$$

The residual and Y_n are jointly Gaussian because they are linear transformations of a Gaussian vector. Their zero cross-covariance implies independence: the Gaussian characteristic function $\exp(-t^\mathsf{T}\Sigma t/2)$ factors across covariance blocks when the cross-covariance block is zero. This argument also allows a zero-variance residual. Since $\mathrm{E}[W_n] = 0$, independence gives $\mathrm{E}[W_n \mid Y_n] = 0$. Taking the conditional expectation in (9) therefore yields

$$\boxed{g_n(y_0, \cdots, y_{n-1}) = \frac{c}{1+(n-1)c}\sum_{j=0}^{n-1} y_j.} \tag{10}$$

This linear function is the unique continuous version, because the nonsingular vector Y_n has a strictly positive density everywhere.

For completeness, the same residual calculation gives

$$\begin{aligned}\operatorname{Var}(X_n \mid Y_n) = \operatorname{Var}(W_n) &= 1 - \frac{nc^2}{1+(n-1)c}\\ &= \frac{(1-c)(1+nc)}{1+(n-1)c}.\end{aligned} \tag{11}$$

This nonnegative conditional variance describes actual test outcomes, not randomness in the recursively specified numbers e_n.

4 Solving the recursion and taking the average

Inserting the prescribed history into (10) gives

$$e_n = \frac{c}{1+(n-1)c}\left(a + \sum_{j=1}^{n-1} e_j\right), \qquad 1 \le n \le k, \tag{12}$$

where the sum is empty at $n = 1$. In particular, $e_1 = ca$. Suppose $e_j = ca$ for all $1 \le j < n$. Then

$$e_n = \frac{c}{1+(n-1)c}[a + (n-1)ca] = \frac{ca[1+(n-1)c]}{1+(n-1)c} = ca.$$

Induction establishes

$$\boxed{e_1 = e_2 = \cdots = e_k = ca.} \tag{13}$$

Consequently the requested average is

$$\boxed{\bar{e}_k = \frac{1}{k}\sum_{n=1}^{k} e_n = ca.} \tag{14}$$

The answer depends on c and on the fixed starting score a, but not on the number k of subsequent predictions, within the admissible domain. Once a and c are fixed, this average is deterministic. An additional expectation changes nothing: $\mathrm{E}[\bar{e}_k] = ca$.

There is also a partial-sum derivation. Define

$$T_0 = a, \qquad T_n = a + \sum_{j=1}^{n} e_j.$$

Equation (12) implies

$$T_n = \frac{1 + nc}{1 + (n-1)c} T_{n-1}.$$

Multiplying these relations telescopes, without dividing by c or by a:

$$T_n = a \prod_{j=1}^{n} \frac{1 + jc}{1 + (j-1)c} = a(1 + nc).$$

Thus $\sum_{j=1}^{k} e_j = T_k - a = kca$, which again gives (14). This proof includes $a = 0$ and $c = 0$.

For example, if $a = 2$ and $c = 0.6$, then

$$e_1 = 1.2, \qquad e_2 = \frac{0.6}{1.6}(2 + 1.2) = 1.2, \qquad e_3 = \frac{0.6}{2.2}(2 + 1.2 + 1.2) = 1.2.$$

Every subsequent prediction and their average remain 1.2.

5 Interpretation and other averages

5.1 Actual future scores

The recursion replaces each previous future score by its conditional mean. Actual observed future scores need not equal those replacements. If

$$M_k = \frac{1}{k}\sum_{n=1}^{k} X_n$$

is the actual future average, bivariate Gaussian regression gives $\mathrm{E}[X_n \mid X_0 = a] = ca$ for every $1 \le n \le k$. Linearity therefore gives

$$\mathrm{E}[M_k \mid X_0 = a] = ca. \tag{15}$$

Although this conditional expectation equals $\bar{e}_k$, the random variable M_k generally has positive conditional variance.

In fact,

$$\mathrm{Var}(M_k) = \frac{1 + (k-1)c}{k}, \qquad \mathrm{Cov}(M_k, X_0) = c,$$

so Gaussian regression gives

$$\mathrm{Var}(M_k \mid X_0 = a) = \frac{1 + (k-1)c}{k} - c^2 = (1-c)\left(c + \frac{1}{k}\right). \tag{16}$$

Likewise, the random predictors

$$P_n = \mathrm{E}[X_n \mid X_0, \dots, X_{n-1}]$$

use actual prior outcomes and are not the deterministic values e_n. The tower property implies $\mathrm{E}[P_n \mid X_0] = cX_0$, while its unconditional implication $\mathrm{E}[P_n] = 0$ holds without Gaussianity whenever the scores are integrable and have mean zero. Applying a conditional mean function at a fixed, recursively chosen history is a different operation.

5.2 Including the starting observation or randomizing it

If the starting value is included, there are $k+1$ terms, and

$$\frac{a + \sum_{n=1}^{k} e_n}{k+1} = \frac{1 + kc}{k+1} a. \tag{17}$$

This differs from the requested average, which excludes a. If one instead randomizes the starting observation, the recursively filled average becomes cX_0. It has unconditional expectation zero. That is a different averaging operation from holding the starting score fixed.

5.3 Returning to a common raw-score scale

For tests with common mean m and standard deviation $s > 0$, put

$$Y_j = m + sX_j.$$

If the fixed raw starting score is b, then $a = (b-m)/s$. Affine equivariance of conditional expectation gives each raw-score prediction and its average as

$$\boxed{e_n^{\mathrm{raw}} = \bar{e}_k^{\mathrm{raw}} = m + sca = m + c(b-m).} \tag{18}$$

Thus s determines the conversion to a standardized starting score; it cancels when the prediction is expressed directly in terms of $b - m$.

6 Boundary cases

At $c = 0$, joint Gaussianity makes the scores independent, and every subsequent prediction is zero. At $c = 1$, $\mathrm{Var}(X_i - X_j) = 2(1-c) = 0$, so all scores agree almost surely. Given the starting score a, every compatible recursive prediction is $a = ca$. This establishes the result at $c = 1$ without using the singular inverse in (7).

At the other endpoint, $c = -1/k$, the full vector satisfies

$$\sum_{j=0}^{k} X_j = 0 \quad \text{almost surely,}$$

because that sum has mean zero and variance $(k+1)(1+kc) = 0$. Every conditioning prefix remains nonsingular, as shown above, so the derivation of (14) applies. Its prescribed history is compatible with the support, since $a + kca = 0$. The final conditional variance (11) is zero.

7 Why correlation alone is insufficient

The original description specifies only marginal means, marginal variances and pairwise correlations. Those quantities do not determine the conditional mean functions. A counterexample already exists with one subsequent test. Let

$$P(X_0 = -\sqrt{2}) = P(X_0 = \sqrt{2}) = \tfrac{1}{4}, \qquad P(X_0 = 0) = \tfrac{1}{2}, \qquad X_1 = X_0^2 - 1.$$

Symmetry and direct calculation give

$$\mathrm{E}[X_0] = 0, \quad \mathrm{E}[X_0^2] = 1, \quad \mathrm{E}[X_0^3] = 0, \quad \mathrm{E}[X_0^4] = 2.$$

Consequently

$$\begin{aligned} \mathrm{E}[X_1] &= \mathrm{E}[X_0^2] - 1 = 0, \\ \mathrm{Var}(X_1) &= \mathrm{E}[(X_0^2 - 1)^2] = 2 - 2 + 1 = 1, \\ \mathrm{Cov}(X_0, X_1) &= \mathrm{E}[X_0^3 - X_0] = 0. \end{aligned}$$

Both scores are standardized and $c = 0$, yet at the fixed starting value $a = 0$, an event with positive probability,

$$e_1 = \mathrm{E}[X_1 \mid X_0 = 0] = -1 \neq 0 = ca.$$

Therefore (14) is a result under the stated joint Gaussian model, or another model with the same conditional-mean functions; it does not follow from equicorrelation alone.

8 Scope and provenance

This analysis extends the repository's hypothetical Gaussian illustration to a sequential prediction question. The Gaussian residual argument and the recursion above provide the mathematical derivation. The repository's existing Ada/SPARK and Python/Deal verification records concern their stated programs and analyses; they are not new formal-proof evidence for this text. No new simulation or formal-prover campaign is claimed here. The accompanying README records how to build the PDF and the checks actually performed for this addition. The repository's original material is governed by its existing license, subject to third-party rights.